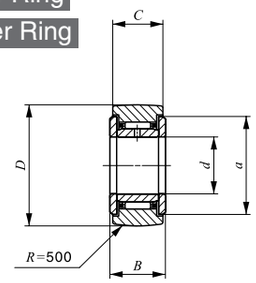
ROLLER FOLLOWERS

Separable Roller Followers, Shield Type **With Cage/With Inner Ring**
Separable Roller Followers, Sealed Type **With Cage/With Inner Ring**





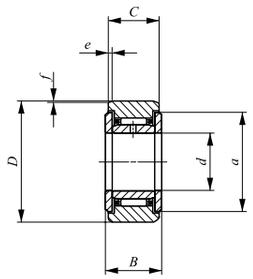


NAST···ZZR

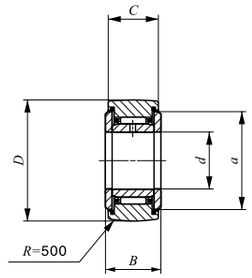
Shaft dia. 6 — 50mm

Shaft dia. mm	Identification number: Shield type, Crowned outer ring	Identification number: Shield type, Cylindrical outer ring	Identification number: Sealed type, Crowned outer ring	Identification number: Sealed type, Cylindrical outer ring	Mass (Ref.) g
6	NAST 6 ZZR	NAST 6 ZZ	NAST 6 ZZUUR	NAST 6 ZZUU	24.5
8	NAST 8 ZZR	NAST 8 ZZ	NAST 8 ZZUUR	NAST 8 ZZUU	39
10	NAST 10 ZZR	NAST 10 ZZ	NAST 10 ZZUUR	NAST 10 ZZUU	65
12	NAST 12 ZZR	NAST 12 ZZ	NAST 12 ZZUUR	NAST 12 ZZUU	75
15	NAST 15 ZZR	NAST 15 ZZ	NAST 15 ZZUUR	NAST 15 ZZUU	83
17	NAST 17 ZZR	NAST 17 ZZ	NAST 17 ZZUUR	NAST 17 ZZUU	135
20	NAST 20 ZZR	NAST 20 ZZ	NAST 20 ZZUUR	NAST 20 ZZUU	195
25	NAST 25 ZZR	NAST 25 ZZ	NAST 25 ZZUUR	NAST 25 ZZUU	225
30	NAST 30 ZZR	NAST 30 ZZ	NAST 30 ZZUUR	NAST 30 ZZUU	400
35	NAST 35 ZZR	NAST 35 ZZ	NAST 35 ZZUUR	NAST 35 ZZUU	550
40	NAST 40 ZZR	NAST 40 ZZ	NAST 40 ZZUUR	NAST 40 ZZUU	710
45	NAST 45 ZZR	NAST 45 ZZ	NAST 45 ZZUUR	NAST 45 ZZUU	760
50	NAST 50 ZZR	NAST 50 ZZ	NAST 50 ZZUUR	NAST 50 ZZUU	830

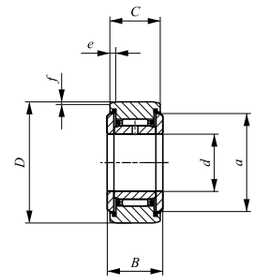
Remarks1. The inner ring has an oil hole.
2. The sealed type is provided with prepacked grease. The shield type is not provided with prepacked grease. Perform proper lubrication for use.



NAST···ZZ



NAST···ZZUUR



NAST···ZZUU

Boundary dimensions mm: d	D	B	C	a	e	f	Basic dynamic load rating C N	Basic static load rating C_0 N
6	19	14	13.8	14	2.5	0.8	4 160	4 550
8	24	14	13.8	17.5	2.5	0.8	5 650	5 890
10	30	16	15.8	23.5	2.5	0.8	9 790	9 680
12	32	16	15.8	25.5	2.5	0.8	10 500	10 900
15	35	16	15.8	29	2.5	0.8	12 400	14 300
17	40	20	19.8	32.5	3	1	17 600	20 900
20	47	20	19.8	38	3	1	19 400	24 500
25	52	20	19.8	43	3	1	20 800	28 400
30	62	25	24.8	50.5	4	1.2	30 500	45 400
35	72	25	24.8	53.5	4	1.2	32 400	50 600
40	80	26	25.8	61.5	4	1.2	35 900	61 100
45	85	26	25.8	66.5	4	1.2	37 400	66 400
50	90	26	25.8	76	4	1.2	38 900	71 700

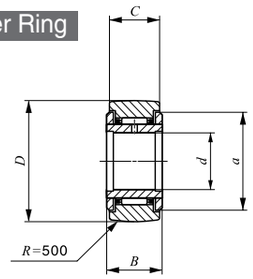
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER FOLLOWERS

Non-separable Roller Followers With Cage/With Inner Ring
Full Complement Type/With Inner Ring





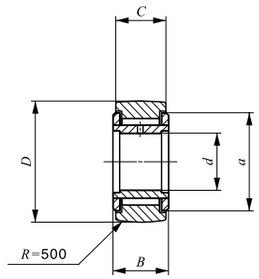


NART···R

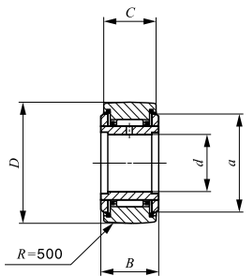
Shaft dia. 5 — 40mm

Shaft dia. mm	Identification number: Shield type, Crowned outer ring, With cage	Identification number: Shield type, Crowned outer ring, Full complement	Identification number: Sealed type, Crowned outer ring, With cage	Identification number: Sealed type, Crowned outer ring, Full complement	Mass (Ref.) g
5	NART 5 R	—	NART 5 UUR	—	14.5
	—	NART 5 VR	—	NART 5 VUUR	15.1
6	NART 6 R	—	NART 6 UUR	—	20.5
	—	NART 6 VR	—	NART 6 VUUR	21.5
8	NART 8 R	—	NART 8 UUR	—	41.5
	—	NART 8 VR	—	NART 8 VUUR	42.5
10	NART 10 R	—	NART 10 UUR	—	64.5
	—	NART 10 VR	—	NART 10 VUUR	66.5
12	NART 12 R	—	NART 12 UUR	—	71
	—	NART 12 VR	—	NART 12 VUUR	73
15	NART 15 R	—	NART 15 UUR	—	102
	—	NART 15 VR	—	NART 15 VUUR	106
17	NART 17 R	—	NART 17 UUR	—	149
	—	NART 17 VR	—	NART 17 VUUR	155
20	NART 20 R	—	NART 20 UUR	—	250
	—	NART 20 VR	—	NART 20 VUUR	255
25	NART 25 R	—	NART 25 UUR	—	285
	—	NART 25 VR	—	NART 25 VUUR	295
30	NART 30 R	—	NART 30 UUR	—	470
	—	NART 30 VR	—	NART 30 VUUR	485
35	NART 35 R	—	NART 35 UUR	—	640
	—	NART 35 VR	—	NART 35 VUUR	655
40	NART 40 R	—	NART 40 UUR	—	845
	—	NART 40 VR	—	NART 40 VUUR	865

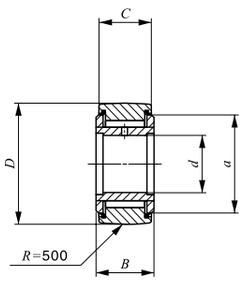
Remarks1. The inner ring has an oil hole.
2. The sealed type is provided with prepacked grease. The shield type is not provided with prepacked grease. Perform proper lubrication for use.



NART···VR



NART···UUR



NART···VUUR

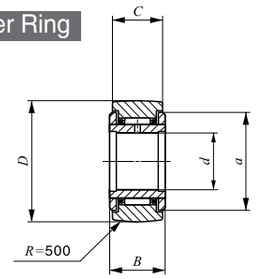
Boundary dimensions mm d	D	B	C	a	Basic dynamic load rating C N	Basic static load rating C_0 N	Maximum allowable static load N
5	16	12	11	12	3 650	3 680	3 680
5	16	12	11	12	6 810	8 370	7 310
6	19	12	11	14	4 250	4 740	4 740
6	19	12	11	14	7 690	10 300	10 300
8	24	15	14	17.5	5 640	5 900	5 900
8	24	15	14	17.5	11 800	15 600	15 600
10	30	15	14	23.5	8 030	7 540	7 540
10	30	15	14	23.5	15 600	18 100	17 500
12	32	15	14	25.5	8 580	8 470	8 470
12	32	15	14	25.5	16 800	20 500	18 600
15	35	19	18	29	13 700	16 400	16 400
15	35	19	18	29	25 200	36 400	24 000
17	40	21	20	32.5	17 600	21 000	21 000
17	40	21	20	32.5	32 000	46 300	33 100
20	47	25	24	38	23 000	30 700	30 700
20	47	25	24	38	41 600	67 300	67 300
25	52	25	24	43	24 700	35 400	35 400
25	52	25	24	43	45 500	79 100	79 100
30	62	29	28	50.5	33 600	51 400	51 400
30	62	29	28	50.5	59 900	110 000	92 500
35	72	29	28	53.5	35 700	57 400	57 400
35	72	29	28	53.5	63 100	121 000	121 000
40	80	32	30	61.5	44 900	81 500	81 500
40	80	32	30	61.5	76 300	164 000	164 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER FOLLOWERS

Non-separable Roller Followers With Cage/With Inner Ring
Full Complement Type/With Inner Ring



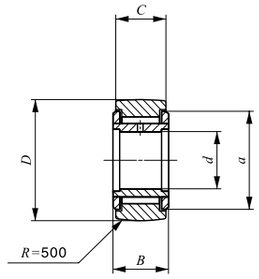


NART···R

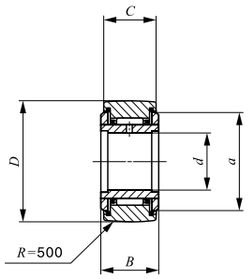
Shaft dia. 45 — 50mm

Shaft dia. mm	Identification number: Shield type, Crowned outer ring, With cage	Identification number: Shield type, Crowned outer ring, Full complement	Identification number: Sealed type, Crowned outer ring, With cage	Identification number: Sealed type, Crowned outer ring, Full complement	Mass (Ref.) g
45	NART 45 R	—	NART 45 UUR	—	915
	—	NART 45 VR	—	NART 45 VUUR	935
50	NART 50 R	—	NART 50 UUR	—	980
	—	NART 50 VR	—	NART 50 VUUR	1 010

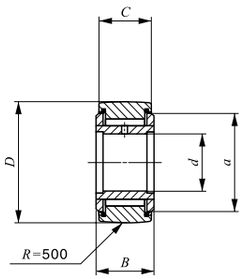
Remarks1. The inner ring has an oil hole.
2. The sealed type is provided with prepacked grease. The shield type is not provided with prepacked grease. Perform proper lubrication for use.



NART···VR



NART···UUR



NART···VUUR

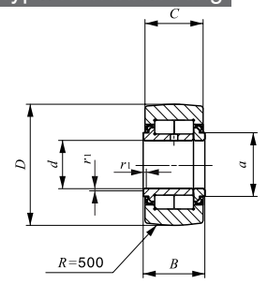
Boundary dimensions mm d	D	B	C	a	Basic dynamic load rating C N	Basic static load rating C_0 N	Maximum allowable static load N
45	85	32	30	66.5	46 800	88 600	88 600
45	85	32	30	66.5	80 300	181 000	181 000
50	90	32	30	76	48 600	95 600	95 600
50	90	32	30	76	84 300	198 000	198 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

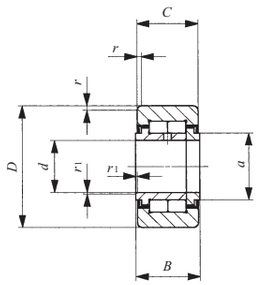
ROLLER FOLLOWERS

Heavy Duty Type Roller Followers Full Complement Type/With Inner Ring





NURT···R



NURT

Shaft dia. 15 — 50mm

Shaft dia. mm	Identification number: Crowned outer ring	Identification number: Cylindrical outer ring	Mass (Ref.) g	Boundary dimensions mm: d	D	B	C	a	$r_{s\,min}$(1)	$r_{1s\,min}$(1)	Basic dynamic load rating C N	Basic static load rating C_0 N	Maximum allowable static load N
15	NURT 15 R	NURT 15	100	15	35	19	18	20	0.6	0.3	23 400	27 300	11 800
	NURT 15-1 R	NURT 15-1	160	15	42	19	18	20	0.6	0.3	23 400	27 300	27 300
17	NURT 17 R	NURT 17	147	17	40	21	20	22	1	0.3	25 200	30 900	20 300
	NURT 17-1 R	NURT 17-1	222	17	47	21	20	22	1	0.3	25 200	30 900	30 900
20	NURT 20 R	NURT 20	245	20	47	25	24	27	1	0.3	38 900	49 000	27 200
	NURT 20-1 R	NURT 20-1	321	20	52	25	24	27	1	0.3	38 900	49 000	49 000
25	NURT 25 R	NURT 25	281	25	52	25	24	31	1	0.3	43 100	58 100	30 000
	NURT 25-1 R	NURT 25-1	450	25	62	25	24	31	1	0.3	43 100	58 100	58 100
30	NURT 30 R	NURT 30	466	30	62	29	28	38	1	0.3	58 200	75 300	35 200
	NURT 30-1 R	NURT 30-1	697	30	72	29	28	38	1	0.3	58 200	75 300	75 300
35	NURT 35 R	NURT 35	630	35	72	29	28	44	1	0.6	63 900	88 800	57 000
	NURT 35-1 R	NURT 35-1	840	35	80	29	28	44	1	0.6	63 900	88 800	88 800
40	NURT 40 R	NURT 40	817	40	80	32	30	49	1	0.6	86 500	122 000	75 300
	NURT 40-1 R	NURT 40-1	1 130	40	90	32	30	49	1	0.6	86 500	122 000	122 000
45	NURT 45 R	NURT 45	883	45	85	32	30	53	1	0.6	91 500	135 000	78 700
	NURT 45-1 R	NURT 45-1	1 400	45	100	32	30	53	1	0.6	91 500	135 000	135 000
50	NURT 50 R	NURT 50	950	50	90	32	30	58	1	0.6	96 300	148 000	82 100
	NURT 50-1 R	NURT 50-1	1 690	50	110	32	30	58	1	0.6	96 300	148 000	148 000

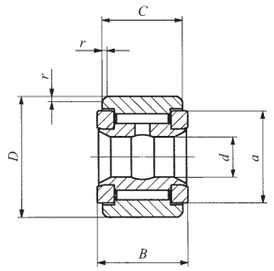
Note(1) Minimum allowable value of chamfer dimension r or r_1
Remarks1. The inner ring has an oil hole.
2. Provided with prepacked grease.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

ROLLER FOLLOWERS

Non-separable Roller Followers, Inch Series Full Complement Type /With Inner Ring



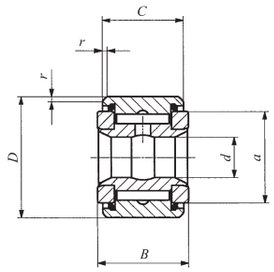


CRY…V

Shaft dia. 6.350 — 31.750mm

Shaft dia. mm (inch)	Identification number: Shield type Cylindrical outer ring	Identification number: Sealed type Cylindrical outer ring	Mass (Ref.) g	Boundary dimensions mm(inch): d	D	B	C
6.350 (1/4)	**CRY 12 V**	**CRY 12 VUU**	27	6.350 (1/4)	19.050 (3/4)	14.288(0.5625)	12.700 (1/2)
	CRY 14 V	**CRY 14 VUU**	36	6.350 (1/4)	22.225 (7/8)	14.288(0.5625)	12.700 (1/2)
7.938 (5/16)	**CRY 16 V**	**CRY 16 VUU**	68	7.938 (5/16)	25.400 (1)	17.463(0.6875)	15.875 (5/8)
	CRY 18 V	**CRY 18 VUU**	77	7.938 (5/16)	28.575 (1 1/8)	17.463(0.6875)	15.875 (5/8)
9.525 (3/8)	**CRY 20 V**	**CRY 20 VUU**	109	9.525 (3/8)	31.750 (1 1/4)	20.638(0.8125)	19.050 (3/4)
	CRY 22 V	**CRY 22 VUU**	136	9.525 (3/8)	34.925 (1 3/8)	20.638(0.8125)	19.050 (3/4)
11.112 (7/16)	**CRY 24 V**	**CRY 24 VUU**	186	11.112 (7/16)	38.100 (1 1/2)	23.813(0.9375)	22.225 (7/8)
	CRY 26 V	**CRY 26 VUU**	227	11.112 (7/16)	41.275 (1 5/8)	23.813(0.9375)	22.225 (7/8)
12.700 (1/2)	**CRY 28 V**	**CRY 28 VUU**	290	12.700 (1/2)	44.450 (1 3/4)	26.988(1.0625)	25.400 (1)
	CRY 30 V	**CRY 30 VUU**	363	12.700 (1/2)	47.625 (1 7/8)	26.988(1.0625)	25.400 (1)
15.875 (5/8)	**CRY 32 V**	**CRY 32 VUU**	476	15.875 (5/8)	50.800 (2)	33.338(1.3125)	31.750 (1 1/4)
	CRY 36 V	**CRY 36 VUU**	599	15.875 (5/8)	57.150 (2 1/4)	33.338(1.3125)	31.750 (1 1/4)
19.050 (3/4)	**CRY 40 V**	**CRY 40 VUU**	816	19.050 (3/4)	63.500 (2 1/2)	39.688(1.5625)	38.100 (1 1/2)
	CRY 44 V	**CRY 44 VUU**	1 020	19.050 (3/4)	69.850 (2 3/4)	39.688(1.5625)	38.100 (1 1/2)
25.400 (1)	**CRY 48 V**	**CRY 48 VUU**	1 410	25.400 (1)	76.200 (3)	46.038(1.8125)	44.450 (1 3/4)
	CRY 52 V	**CRY 52 VUU**	1 640	25.400 (1)	82.550 (3 1/4)	46.038(1.8125)	44.450 (1 3/4)
28.575 (1 1/8)	**CRY 56 V**	**CRY 56 VUU**	2 250	28.575 (1 1/8)	88.900 (3 1/2)	52.388(2.0625)	50.800 (2)
31.750 (1 1/4)	**CRY 64 V**	**CRY 64 VUU**	3 200	31.750 (1 1/4)	101.600 (4)	58.738(2.3125)	57.150 (2 1/4)

Remarks1. The inner ring has an oil groove and an oil hole.
2. Provided with prepacked grease.



CRY…VUU

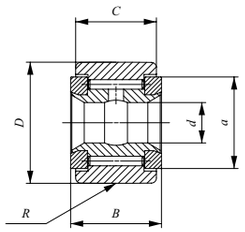
a	r	Shaft dia. Push fit Min.	Shaft dia. Push fit Max.	Shaft dia. Drive fit Min.	Shaft dia. Drive fit Max.	Shaft dia. Press fit Min.	Shaft dia. Press fit Max.	Basic dynamic load rating C N	Basic static load rating C_0 N
14.4(0.567)	0.794 (1/32)	6.332	6.342	6.348	6.358	6.353	6.363	8 710	12 300
14.4(0.567)	0.794 (1/32)	6.332	6.342	6.348	6.358	6.353	6.363	8 710	12 300
19.6(0.772)	1.191 (3/64)	7.920	7.930	7.935	7.945	7.940	7.950	13 100	22 700
19.6(0.772)	1.588 (1/16)	7.920	7.930	7.935	7.945	7.940	7.950	13 100	22 700
25.0(0.984)	1.588 (1/16)	9.507	9.517	9.523	9.533	9.528	9.538	23 600	31 700
25.0(0.984)	1.588 (1/16)	9.507	9.517	9.523	9.533	9.528	9.538	23 600	31 700
28.8(1.134)	1.588 (1/16)	11.095	11.105	11.110	11.120	11.115	11.125	28 200	40 100
28.8(1.134)	1.588 (1/16)	11.095	11.105	11.110	11.120	11.115	11.125	28 200	40 100
32.7(1.287)	1.588 (1/16)	12.682	12.692	12.698	12.708	12.708	12.718	35 300	55 600
32.7(1.287)	1.588 (1/16)	12.682	12.692	12.698	12.708	12.708	12.718	35 300	55 600
36.0(1.417)	1.588 (1/16)	15.857	15.867	15.873	15.883	15.883	15.893	45 700	80 600
36.0(1.417)	1.588 (1/16)	15.857	15.867	15.873	15.883	15.883	15.893	45 700	80 600
43.3(1.705)	2.381 (3/32)	19.032	19.042	19.048	19.058	19.058	19.068	61 400	116 000
43.3(1.705)	2.381 (3/32)	19.032	19.042	19.048	19.058	19.058	19.068	61 400	116 000
54.0(2.125)	2.381 (3/32)	25.377	25.390	25.397	25.410	25.408	25.420	77 600	172 000
54.0(2.125)	2.381 (3/32)	25.377	25.390	25.397	25.410	25.408	25.420	77 600	172 000
61.9(2.437)	2.381 (3/32)	28.522	28.565	28.572	28.585	28.583	28.595	111 000	239 000
71.0(2.797)	2.381 (3/32)	31.727	31.740	31.747	31.760	31.758	31.770	142 000	317 000

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

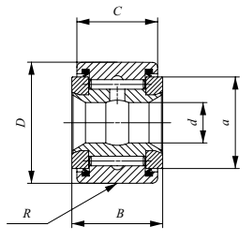
ROLLER FOLLOWERS

Non-separable Roller Followers, Inch Series **Full Complement Type / With Inner Ring**





CRY…VR



CRY…VUUR

Shaft dia. 6.350 — 31.750mm

Shaft dia. mm (inch)	Identification number: Shield type Crowner outer ring	Identification number: Sealed type Crowned outer ring	Mass (Ref.) g	Boundary dimensions mm(inch): d	D	B	C
6.350 (1/4)	CRY 12 VR	CRY 12 VUUR	27	6.350 (1/4)	19.050 (3/4)	14.288(0.5625)	12.700 (1/2)
	CRY 14 VR	CRY 14 VUUR	36	6.350 (1/4)	22.225 (7/8)	14.288(0.5625)	12.700 (1/2)
7.938 (5/16)	CRY 16 VR	CRY 16 VUUR	68	7.938 (5/16)	25.400 (1)	17.463(0.6875)	15.875 (5/8)
	CRY 18 VR	CRY 18 VUUR	77	7.938 (5/16)	28.575 (1 1/8)	17.463(0.6875)	15.875 (5/8)
9.525 (3/8)	CRY 20 VR	CRY 20 VUUR	109	9.525 (3/8)	31.750 (1 1/4)	20.638(0.8125)	19.050 (3/4)
	CRY 22 VR	CRY 22 VUUR	136	9.525 (3/8)	34.925 (1 3/8)	20.638(0.8125)	19.050 (3/4)
11.112 (7/16)	CRY 24 VR	CRY 24 VUUR	186	11.112 (7/16)	38.100 (1 1/2)	23.813(0.9375)	22.225 (7/8)
	CRY 26 VR	CRY 26 VUUR	227	11.112 (7/16)	41.275 (1 5/8)	23.813(0.9375)	22.225 (7/8)
12.700 (1/2)	CRY 28 VR	CRY 28 VUUR	290	12.700 (1/2)	44.450 (1 3/4)	26.988(1.0625)	25.400 (1)
	CRY 30 VR	CRY 30 VUUR	363	12.700 (1/2)	47.625 (1 7/8)	26.988(1.0625)	25.400 (1)
15.875 (5/8)	CRY 32 VR	CRY 32 VUUR	476	15.875 (5/8)	50.800 (2)	33.338(1.3125)	31.750 (1 1/4)
	CRY 36 VR	CRY 36 VUUR	599	15.875 (5/8)	57.150 (2 1/4)	33.338(1.3125)	31.750 (1 1/4)
19.050 (3/4)	CRY 40 VR	CRY 40 VUUR	816	19.050 (3/4)	63.500 (2 1/2)	39.688(1.5625)	38.100 (1 1/2)
	CRY 44 VR	CRY 44 VUUR	1 020	19.050 (3/4)	69.850 (2 3/4)	39.688(1.5625)	38.100 (1 1/2)
25.400 (1)	CRY 48 VR	CRY 48 VUUR	1 410	25.400 (1)	76.200 (3)	46.038(1.8125)	44.450 (1 3/4)
	CRY 52 VR	CRY 52 VUUR	1 640	25.400 (1)	82.550 (3 1/4)	46.038(1.8125)	44.450 (1 3/4)
28.575 (1 1/8)	CRY 56 VR	CRY 56 VUUR	2 250	28.575 (1 1/8)	88.900 (3 1/2)	52.388(2.0625)	50.800 (2)
31.750 (1 1/4)	CRY 64 VR	CRY 64 VUUR	3 200	31.750 (1 1/4)	101.600 (4)	58.738(2.3125)	57.150 (2 1/4)

a	R	Shaft dia. Push fit Min.	Push fit Max.	Drive fit Min.	Drive fit Max.	Press fit Min.	Press fit Max.	Basic dynamic load rating C N	Basic static load rating C_0 N
14.4(0.567)	250 (10)	6.332	6.342	6.348	6.358	6.353	6.363	8 710	12 300
14.4(0.567)	250 (10)	6.332	6.342	6.348	6.358	6.353	6.363	8 710	12 300
19.6(0.772)	300 (12)	7.920	7.930	7.935	7.945	7.940	7.950	13 100	22 700
19.6(0.772)	300 (12)	7.920	7.930	7.935	7.945	7.940	7.950	13 100	22 700
25.0(0.984)	360 (14)	9.507	9.517	9.523	9.533	9.528	9.538	23 600	31 700
25.0(0.984)	360 (14)	9.507	9.517	9.523	9.533	9.528	9.538	23 600	31 700
28.8(1.134)	500 (20)	11.095	11.105	11.110	11.120	11.115	11.125	28 200	40 100
28.8(1.134)	500 (20)	11.095	11.105	11.110	11.120	11.115	11.125	28 200	40 100
32.7(1.287)	500 (20)	12.682	12.692	12.698	12.708	12.708	12.718	35 300	55 600
32.7(1.287)	500 (20)	12.682	12.692	12.698	12.708	12.708	12.718	35 300	55 600
36.0(1.417)	600 (24)	15.857	15.867	15.873	15.883	15.883	15.893	45 700	80 600
36.0(1.417)	600 (24)	15.857	15.867	15.873	15.883	15.883	15.893	45 700	80 600
43.3(1.705)	760 (30)	19.032	19.042	19.048	19.058	19.058	19.068	61 400	116 000
43.3(1.705)	760 (30)	19.032	19.042	19.048	19.058	19.058	19.068	61 400	116 000
54.0(2.125)	760 (30)	25.377	25.390	25.397	25.410	25.408	25.420	77 600	172 000
54.0(2.125)	760 (30)	25.377	25.390	25.397	25.410	25.408	25.420	77 600	172 000
61.9(2.437)	760 (30)	28.522	28.565	28.572	28.585	28.583	28.595	111 000	239 000
71.0(2.797)	760 (30)	31.727	31.740	31.747	31.760	31.758	31.770	142 000	317 000

Remarks1. The inner ring has an oil groove and an oil hole.
2. Provided with prepacked grease.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CROSSED ROLLER BEARINGS

- ●Mounting Holed Type High Rigidity Crossed Roller Bearing
- ●High Rigidity Type Crossed Roller Bearings
- ●Standard Type Crossed Roller Bearings
- ●Slim Type Crossed Roller Bearings



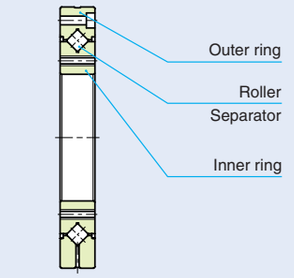
Structure and Features

IKO Crossed Roller Bearings are compact bearings with their rollers alternately crossed at right angles to each other between inner and outer rings. They can take loads from any directions at the same time such as radial, thrust and moment loads. The rollers make line-contact with raceway surfaces, and, therefore, elastic deformation due to bearing loads is very small. These bearings are widely used in the rotating parts of industrial robots, machine tools, medical equipment, etc., which require compactness, high rigidity and high rotational accuracy.

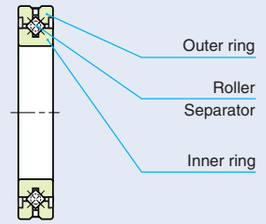
In addition, bearings made of stainless steel or those with inner and outer rings provided with mounting holes are also available on request. Please contact IKO.

Structure of Crossed Roller Bearings

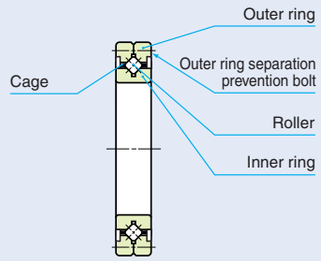
CRBF···A



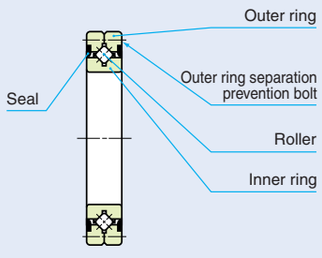
CRBH···A



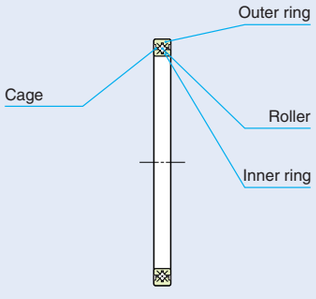
CRBC



CRB···UU



CRBS



J

CRBF
CRBH
CRBC
CRB
CRBS

Types

Crossed Roller Bearings are available in the types shown in Table 1.

Table 1 Crossed Roller Bearing Type

Type		With Cage	With Separator	Full complement
Mounting holed type high rigidity crossed roller bearing CRBF	Open type	—	CRBF ··· A	—
	Sealed type	—	CRBF ··· AUU	—
High rigidity type crossed roller bearings CRBH	Open type	—	CRBH ··· A	—
	Sealed type	—	CRBH ··· AUU	—
Standard type crossed roller bearings CRBC, CRB	Open type	CRBC	—	CRB
	Sealed type	CRBC ··· UU	—	CRB ··· UU
Slim type crossed roller bearings CRBS	Open type	CRBS	—	CRBS ··· V
	Sealed type	—	CRBS ··· AUU	CRBS ··· VUU

Mounting Holed Type High Rigidity Crossed Roller Bearing

Mounting holes are prepared on outer ring and inner ring providing easy mounting together with high rigidity and high accuracy.

High Rigidity Type Crossed Roller Bearings

Both inner and outer rings have a solid one-piece construction. Therefore, high accuracy and high rigidity are achieved, and mounting errors can be minimized. As separators are incorporated between the rollers for smooth rotation, these bearings are suitable for applications where rotational speed is comparatively high.

Standard Type Crossed Roller Bearings

The outer ring is made of two split pieces, which are bolted together to prevent separation during transportation or mounting. So, handling is easy.

Slim Type Crossed Roller Bearings

These bearings are very slim bearings having a small outside diameter, in comparison with the bore diameter, and a narrow width. The type with cage and the type with separator provide smooth rotation and are suitable for applications where rotational speed is comparatively high.

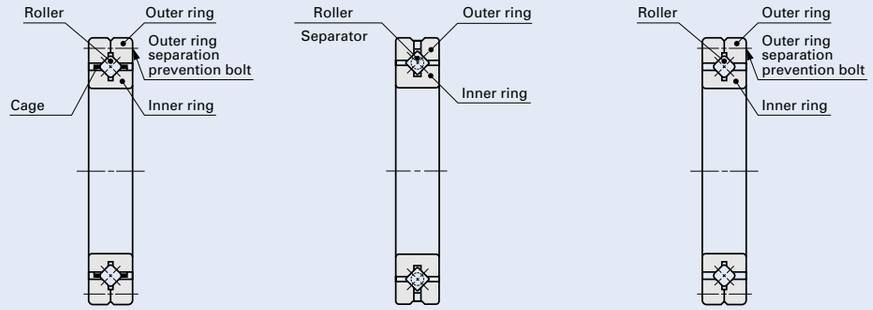
Internal Structures and Shapes

Various types are lined up in Crossed Roller Bearing series, including the type with cage, the type with separator, open type, sealed type, etc..

Roller guide method

Crossed Roller Bearings include the type with cage, type with separator and full complement type. The type with cage and the type with separator have a small coefficient of friction and are suitable for comparatively high speed rotations, while the full complement type is suitable for heavy load applications at low speed rotations.

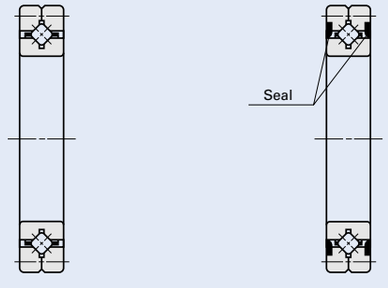
《With Cage》 《With Separator》 《Full complement》



Seal structure

Crossed Roller Bearings include the open type and sealed type. The sealed type bearing incorporates seals made of special synthetic rubber that have excellent sealing performance against dust and dirt penetration and grease leakage.

《Open type》 《Sealed type》

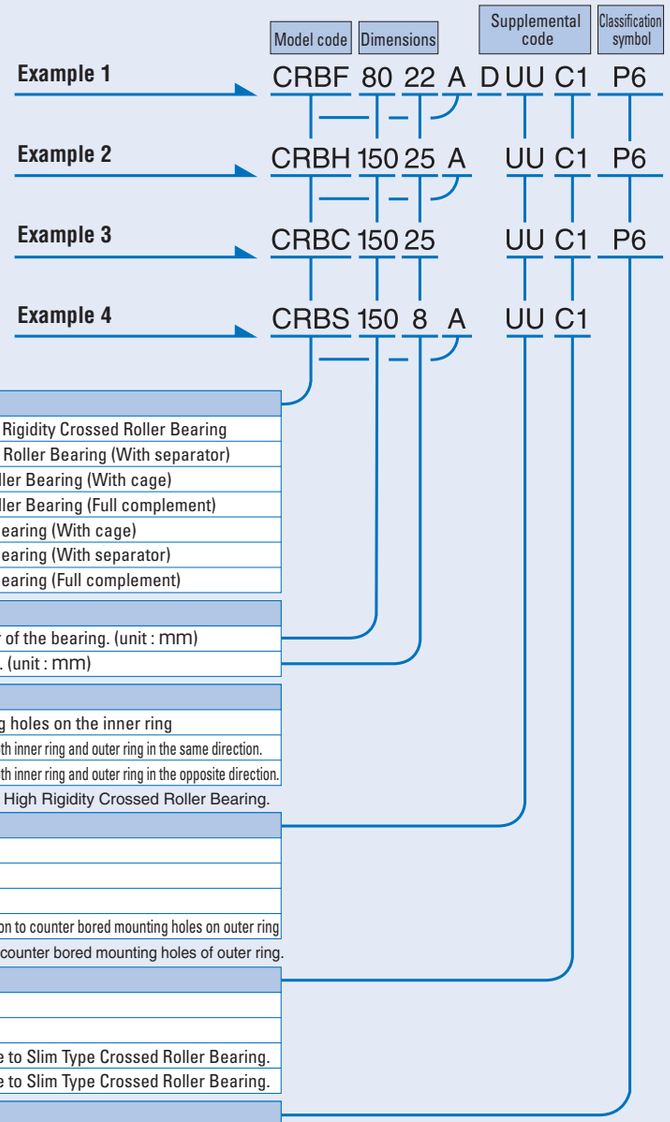


1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Identification number

The identification number of Crossed Roller Bearings consists of a model code, dimensions, any supplemental codes and a classification symbol. Some examples are shown below.

Examples of identification number



Model code	
CRBF···A	Mounting Holed Type High Rigidity Crossed Roller Bearing
CRBH···A	High Rigidity Type Crossed Roller Bearing (With separator)
CRBC	Standard Type Crossed Roller Bearing (With cage)
CRB	Standard Type Crossed Roller Bearing (Full complement)
CRBS	Slim Type Crossed Roller Bearing (With cage)
CRBS···A	Slim Type Crossed Roller Bearing (With separator)
CRBS···V	Slim Type Crossed Roller Bearing (Full complement)

Dimension
The dimension indicates the bore diameter of the bearing. (unit : mm)
The dimension indicates the bearing width. (unit : mm)

Supplemental code - 1(1)	
T	With female threaded mounting holes on the inner ring
No symbol	With counter-bored mounting holes on both inner ring and outer ring in the same direction.
D	With counter-bored mounting holes on both inner ring and outer ring in the opposite direction.

Note(1) Applicable to Mounting Holed Type High Rigidity Crossed Roller Bearing.

Supplemental code - 2	
No symbol	Open type
UU	Sealed type
U	One side sealed type
UD	One side sealed in the opposite direction to counter bored mounting holes on outer ring

Note(2) For CRBF···A, sealed at the side with counter bored mounting holes of outer ring.

Supplemental code - 3		
T1	: T1 clearance	
C1	: C1 clearance	
C2	: C2 clearance	Not applicable to Slim Type Crossed Roller Bearing.
No symbol	:Normal clearance	Not applicable to Slim Type Crossed Roller Bearing.

Classification symbol		
No symbol	Accuracy class 0	
P6	Accuracy class 6	Not applicable to Slim Type Crossed Roller Bearings.
P5	Accuracy class 5	
P4	Accuracy class 4	
P2	Accuracy class 2	

Dynamic Equivalent Load

The dynamic equivalent radial load of Crossed Roller Bearings can be obtained from the following equation.

$$P_r = X\left(F_r + \frac{2M}{D_{pw}}\right) + YF_a \quad \cdots\cdots(1)$$

where, P_r : Dynamic equivalent radial load, N
F_r : Radial load, N
F_a : Axial load, N
M : Moment, N-mm
D_{pw} : Pitch circle diameter of roller set, mm

$$\left(D_{pw} \fallingdotseq \frac{d+D}{2}\right)$$

X : Radial load factor (Refer to Table 2.)
Y : Axial load factor (Refer to Table 2.)

Static Equivalent Load

The static equivalent radial load of Crossed Roller Bearings can be obtained from the following equation.

$$P_{0r} = F_r + \frac{2M}{D_{pw}} + 0.44\,F_a \quad \cdots\cdots(2)$$

where, P_{0r} : Static equivalent radial load, N
F_r : Radial load, N
F_a : Axial load, N
M : Moment, N-mm
D_{pw} : Pitch circle diameter of roller set, mm

$$\left(D_{pw} \fallingdotseq \frac{d+D}{2}\right)$$

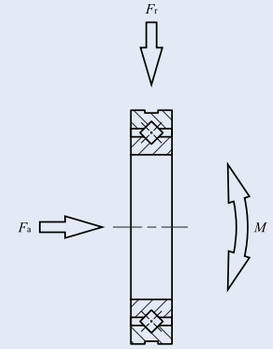


Fig. 1 Load direction

Table 2 Radial load factor and axial load factor

Conditions	X	Y
$\frac{F_a}{F_r + 2M/D_{pw}} \leqq 1.5$	1	0.45
$\frac{F_a}{F_r + 2M/D_{pw}} > 1.5$	0.67	0.67

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Accuracy

The accuracy of Crossed Roller Bearings is shown in Tables 3 and 4. However, the accuracy of Slim Type Crossed Roller Bearings is based on Table 5.

Bearings with special accuracy are also optionally available. Please consult IKO.

Table 3 Tolerances and allowable values of inner rings and tolerances of outer ring width

unit: μm

d Nominal bore diameter mm		Δ_{dmp} (1) Single plane mean bore dia. deviation								Δ_{Bs} Deviation of a single inner ring width		Δ_{Cs} (2) Deviation of a single outer ring width		K_{ia} Radial run-out of assembled bearing inner ring					S_{ia} Assembled bearing inner ring face run-out with raceway				
		Class 0		Class 6		Class 5		Class 4						Class	Class	Class	Class	Class	Class	Class	Class	Class	Class
Over	Incl.	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	0	6	5	4	2	0	6	5	4	2
18	30	0	−10	0	− 8	0	− 6	0	− 5	0	− 75	0	−100	13	8	4	3	2.5	13	8	4	3	2.5
30	50	0	−12	0	−10	0	− 8	0	− 6	0	− 75	0	−100	15	10	5	4	2.5	15	10	5	4	2.5
50	80	0	−15	0	−12	0	− 9	0	− 7	0	− 75	0	−100	20	10	5	4	2.5	20	10	5	4	2.5
80	120	0	−20	0	−15	0	−10	0	− 8	0	− 75	0	−100	25	13	6	5	2.5	25	13	6	5	2.5
120	150	0	−25	0	−18	0	−13	0	−10	0	−100	0	−120	30	18	8	6	2.5	30	18	8	6	2.5
150	180	0	−25	0	−18	0	−13	0	−10	0	−100	0	−120	30	18	8	6	5	30	18	8	6	5
180	250	0	−30	0	−22	0	−15	0	−12	0	−100	0	−120	40	20	10	8	5	40	20	10	8	5
250	315	0	−35	0	−25	0	−18	—	—	0	−120	0	−150	50	25	13	10	7	50	25	13	10	7
315	400	0	−40	0	−30	0	−23	—	—	0	−150	0	−200	60	30	15	12	8	60	30	15	12	8
400	500	0	−45	0	−35	—	—	—	—	0	−150	0	−200	65	35	18	14	10	65	35	18	14	10
500	630	0	−50	0	−40	—	—	—	—	0	−150	0	−200	70	40	20	16	12	70	40	20	16	12
630	800	0	−75	—	—	—	—	—	—	0	−150	0	−200	80	50	25	20	15	80	50	25	20	15

Notes(1) When values are not indicated in the table (Class 2, etc.), those for the highest class for which the values are indicated are applicable.
(2) In case of High Rigidity Type Crossed Roller Bearings, the tolerances for deviation of a single inner ring width are applicable to those of a single outer ring width.
Remark The accuracy specified in this table is not applicable to Slim Type Crossed Roller Bearings.

Table 4 Tolerances and allowable values of outer ring

unit: μm

D Nominal outside diameter mm		Δ_{dmp} (1) Single plane mean outside dia. deviation								K_{ea} Radial run-out of assembled bearing outer ring					S_{ea} Assembled bearing outer ring face run-out with raceway				
		Class 0		Class 6		Class 5		Class 4		Class	Class	Class	Class	Class	Class	Class	Class	Class	Class
Over	Incl.	High	Low	High	Low	High	Low	High	Low	0	6	5	4(2)	2(2)	0	6	5	4(2)	2(2)
30	50	0	− 11	0	− 9	0	− 7	0	− 6	20	10	7	5	2.5	20	10	7	5	2.5
50	80	0	− 13	0	−11	0	− 9	0	− 7	25	13	8	5	4	25	13	8	5	4
80	120	0	− 15	0	−13	0	−10	0	− 8	35	18	10	6	5	35	18	10	6	5
120	150	0	− 18	0	−15	0	−11	0	− 9	40	20	11	7	5	40	20	11	7	5
150	180	0	− 25	0	−18	0	−13	0	−10	45	23	13	8	5	45	23	13	8	5
180	250	0	− 30	0	−20	0	−15	0	−11	50	25	15	10	7	50	25	15	10	7
250	315	0	− 35	0	−25	0	−18	0	−13	60	30	18	11	7	60	30	18	11	7
315	400	0	− 40	0	−28	0	−20	—	—	70	35	20	—	—	70	35	20	—	—
400	500	0	− 45	0	−33	0	−23	—	—	80	40	23	—	—	80	40	23	—	—
500	630	0	− 50	0	−38	0	−28	—	—	100	50	25	—	—	100	50	25	—	—
630	800	0	− 75	0	−45	—	—	—	—	120	60	30	—	—	120	60	30	—	—
800	1000	0	−100	0	−60	—	—	—	—	120	75	35	—	—	120	75	35	—	—
1000	1030	0	−125	—	—	—	—	—	—	120	75	35	—	—	120	75	35	—	—

Notes(1) When values are not indicated in the table (Class 2, etc.), those for the highest class for which the values are indicated are applicable.
(2) Classes 4 and 2 apply to High Rigidity Type Crossed Roller Bearings. For Standard Type Crossed Roller Bearings, the tolerance values for Class 5 are applicable to Classes 4 and 2.
Remark The accuracy specified in this table is not applicable to Slim Type Crossed Roller Bearings.

Table 5 Tolerances and allowable values of inner rings of Mounting Holed Type High Rigidity Crossed Roller Bearing

unit: μm

d Nominal bore diameter mm		Δ_{dmp} (1) Single plane mean bore dia. deviation								Δ_{Bs} Deviation of a single inner ring width		K_{ia} Radial run-out of assembled bearing inner ring					S_{ia} Assembled bearing inner ring face run-out with raceway				
		Class 0		Class 6		Class 5		Class 4				Class	Class	Class	Class	Class	Class	Class	Class	Class	Class
Over	Incl.	High	Low	High	Low	High	Low	High	Low	High	Low	0	6	5	4	2	0	6	5	4	2
—	20	0	− 10	0	− 8	0	− 6	0	− 5	0	− 75	13	8	4	3	2.5	13	8	4	3	2.5
20	30	0	− 10	0	− 8	0	− 6	0	− 5	0	− 75	15	10	5	4	2.5	15	10	5	4	2.5
30	35	0	− 12	0	− 10	0	− 8	0	− 6	0	− 75	15	10	5	4	2.5	15	10	5	4	2.5
35	50	0	− 12	0	− 10	0	− 8	0	− 6	0	− 75	20	10	5	4	2.5	20	10	5	4	2.5
50	65	0	− 15	0	− 12	0	− 9	0	− 7	0	− 75	20	10	5	4	2.5	20	10	5	4	2.5
65	80	0	− 15	0	− 12	0	− 9	0	− 7	0	− 75	25	13	6	5	2.5	25	13	6	5	2.5

Table 6 Tolerances and allowable values of outer rings of Mounting Holed Type High Rigidity Crossed Roller Bearing

unit: μm

d Nominal bore diameter mm		Δ_{dmp} (1) Single plane mean bore dia. deviation								Δ_{Cs} Deviation of a single outer ring width		K_{ia} Radial run-out of assembled bearing inner ring					S_{ia} Assembled bearing inner ring face run-out with raceway				
		Class 0		Class 6		Class 5		Class 4				Class	Class	Class	Class	Class	Class	Class	Class	Class	Class
Over	Incl.	High	Low	High	Low	High	Low	High	Low	High	Low	0	6	5	4	2	0	6	5	4	2
50	60	0	− 13	0	− 11	0	− 9	0	− 7	0	− 75	20	10	7	5	2.5	20	10	7	5	2.5
60	80	0	− 13	0	− 11	0	− 9	0	− 7	0	− 75	25	13	8	5	4	25	13	8	5	4
80	95	0	− 15	0	− 13	0	− 10	0	− 8	0	− 75	25	13	8	5	4	25	13	8	5	4
95	120	0	− 15	0	− 13	0	− 10	0	− 8	0	− 75	35	18	10	6	5	35	18	10	6	5
120	140	0	− 18	0	− 15	0	− 11	0	− 9	0	− 75	35	18	10	6	5	35	18	10	6	5
140	150	0	− 18	0	− 15	0	− 11	0	− 9	0	− 75	40	20	11	7	5	40	20	11	7	5
150	165	0	− 25	0	− 18	0	− 13	0	− 10	0	− 75	40	20	11	7	5	40	20	11	7	5

Table 7 Tolerances and allowable values of Slim Type Crossed Roller Bearings

unit: μm

d Nominal bore diameter mm	Δ_{dmp} Single plane mean bore dia. deviation		Δ_{Dmp} Single plane mean outside dia. deviation		Δ_{Bs} and Δ_{Cs} Deviations of a single inner ring width and outer ring width		K_{ia} and S_{ia} Radial and axial run-out of assembled bearing inner ring	K_{ea} and S_{ea} Radial and axial run-out of assembled bearing outer ring
	High	Low	High	Low	High	Low		
50	0	− 15	0	− 13	0	− 127	13	13
60	0	− 15	0	− 13	0	− 127	13	13
70	0	− 15	0	− 15	0	− 127	15	15
80	0	− 20	0	− 15	0	− 127	15	15
90	0	− 20	0	− 15	0	− 127	15	15
100	0	− 20	0	− 15	0	− 127	15	15
110	0	− 20	0	− 20	0	− 127	20	20
120	0	− 25	0	− 20	0	− 127	20	20
130	0	− 25	0	− 25	0	− 127	25	25
140	0	− 25	0	− 25	0	− 127	25	25
150	0	− 25	0	− 25	0	− 127	25	25
160	0	− 25	0	− 25	0	− 127	25	25
170	0	− 25	0	− 30	0	− 127	25	25
180	0	− 30	0	− 30	0	− 127	30	30
190	0	− 30	0	− 30	0	− 127	30	30
200	0	− 30	0	− 30	0	− 127	30	30

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Clearance

The radial internal clearances of Crossed Roller Bearings are shown in Table 8.1. However, the radial internal clearances of Slim Type Crossed Roller Bearings are based on Table 8.2 and Mounting Holed Type High Rigidity Crossed Roller Bearings are based on Table 8.3.

Table 8.1 Radial internal clearances unit: μm

d Nominal bore diameter mm		Radial internal clearance					
		T1		C1		C2	
Over	Incl.	Min.	Max.	Min.	Max.	Min.	Max.
—	30	−10	0	0	10	10	20
30	40	−10	0	0	10	10	20
40	50	−10	0	0	10	10	25
50	65	−10	0	0	10	10	25
65	80	−10	0	0	15	15	30
80	100	−10	0	0	15	15	35
100	120	−15	0	0	15	15	35
120	140	−15	0	0	20	20	45
140	160	−15	0	0	20	20	50
160	200	−15	0	0	20	20	50
200	250	−20	0	0	25	25	60
250	315	−20	0	0	25	25	60
315	400	−25	0	0	30	30	70
400	500	−30	0	0	40	40	85
500	630	−30	0	0	50	50	100
630	710	−30	0	0	60	60	120
710	800	−40	0	0	70	70	140

Remark This table is not applicable to Slim Type Crossed Roller Bearings.

Table 8.2 Radial internal clearances of Slim Type Crossed Roller Bearings unit: μm

d Nominal bore diameter mm	Radial internal clearance					
	T1		C1		Normal	
	Min.	Max.	Min.	Max.	Min.	Max.
50	−8	0	0	15	30	56
60	−8	0	0	15	30	56
70	−8	0	0	15	30	56
80	−8	0	0	15	41	66
90	−8	0	0	15	41	66
100	−8	0	0	15	41	66
110	−8	0	0	15	41	66
120	−8	0	0	15	51	76
130	−8	0	0	15	51	76
140	−8	0	0	15	51	76
150	−8	0	0	15	51	76
160	−10	0	0	20	51	76
170	−10	0	0	20	51	76
180	−10	0	0	20	61	86
190	−10	0	0	20	61	86
200	−10	0	0	20	61	86

Table 8.3 Radial internal clearances of Mounting Holed Type High Rigidity Crossed Roller Bearing unit: μm

d Nominal bore diameter mm		Radial internal clearance					
		T1		C1		C2	
Over	Incl.	Min.	Max.	Min.	Max.	Min.	Max.
—	20	−10	0	0	10	10	20
20	25	−10	0	0	10	10	20
25	35	−10	0	0	10	10	25
35	65	−10	0	0	15	15	30
65	80	−10	0	0	15	15	35

Fit

The standard fits of Crossed Roller Bearings are shown in Table 9.1. For large bearings, fit based on the actual measured dimensions of the bearings is recommended, and fit allowance should be chosen as small as possible in accordance with the tolerance class given in Table 9.1. When complex loads or shock loads are applied or when high rotational accuracy and rigidity of the bearing are required, it is recommended to use a slight interference fit adjusted to the actual measured dimensions for both inner and outer rings.

For the interference fit, the radial internal clearance after the fit decreases by approximately 70% to 90% of the interference amount. To avoid excessive preload due to fit, it is recommended to use a slight interference fit adjusted to the actual measured dimensions for both T1 and C1 clearances.

Allowable rotational speed

Allowable rotational speeds of Crossed Roller Bearings are affected by mounting and operating conditions. The values in general operation are shown in Table 10.

Table 10 $d_m n$ values(1) of Crossed Roller Bearings

Type \ Lubricant		Grease	Oil
With cage or with separator	Open type	75 000	150 000
	Closed type	60 000	—
Full complement	Open type	50 000	75 000
	Closed type	40 000	—

Note(1) · $d_m n$ value $= d_m \times n$
where, d_m : Mean value of bearing bore and outside diameters, mm
n : Number of rotations per minute, rpm

Table 9.1 Recommended fits for Crossed Roller Bearings under normal load

Radial internal clearance	Tolerance class			
	Inner ring rotating load		Outer ring rotating load	
	Shaft	Housing bore	Shaft	Housing bore
C1 clearance	h5	H7	g5	J7(1)
C2 clearance	j5	H7	g5	J7(1)

Note(1) It is recommended that a slight interference fit adjusted to the actual measured dimensions of the bearing is used.

Table 9.2 Recommended fits for Slim Type Crossed Roller Bearings with normal clearances
(Dimensional tolerances of shaft and housing bore) unit: μm

d Nominal bore diameter mm	Inner ring rotating load				Outer ring rotating load			
	Shaft		Housing bore		Shaft		Housing bore	
	High	Low	High	Low	High	Low	High	Low
50	+15	0	+13	0	−15	−30	−13	−25
60	+15	0	+13	0	−15	−30	−13	−25
70	+15	0	+15	0	−15	−30	−15	−30
80	+20	0	+15	0	−20	−40	−15	−30
90	+20	0	+15	0	−20	−40	−15	−30
100	+20	0	+15	0	−20	−40	−15	−30
110	+20	0	+20	0	−20	−40	−20	−40
120	+25	0	+20	0	−25	−50	−20	−40
130	+25	0	+25	0	−25	−50	−25	−50
140	+25	0	+25	0	−25	−50	−25	−50
150	+25	0	+25	0	−25	−50	−25	−50
160	+25	0	+25	0	−25	−50	−25	−50
170	+25	0	+30	0	−25	−50	−30	−60
180	+30	0	+30	0	−30	−60	−30	−60
190	+30	0	+30	0	−30	−60	−30	−60
200	+30	0	+30	0	−30	−60	−30	−60

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

Lubrication

These bearings are generally lubricated with grease. Grease is supplied through the clearance between the inner ring and the outer ring.
In the sealed type bearings, ALVANIA GREASE EP2 is prepacked as the lubricating grease.
For bearings without prepacked grease, supply grease or oil for use. Operating without grease or oil will increase the wear of the rolling contact surfaces and cause a short bearing life.
When using a special grease, carefully examine the grease properties and contents such as base oil viscosity and extreme pressure additives. In this case, please contact IKO.

Oil Hole

For Crossed Roller Bearings, oil holes and oil grooves can be provided on bearing rings on request. When an oil hole is required on the outer ring, attach "-OH" before the clearance symbol in the identification number. When an oil hole and an oil groove are required on the outer ring, attach "-OG" at the same place in the identification number. For an oil hole on the inner ring, attach "/OH", and for an oil hole and an oil groove on the inner ring, attach "/OG", at the same place in the identification number. High Rigidity Type Crossed Roller Bearings have an oil groove and two oil holes on the outer ring as standard. Table 9 shows availability of oil holes for each bearing type.

Table 9 Oil holes

Bearing type (1)	Oil hole code			
	/nOH	/nOG	-nOH	-nOG
CRBH	○	○	—	— (2)
CRB、CRBC	○	○	○	○
CRBS	○	—	○	—

Notes(1) Only representative types are shown in the table, but this table is applicable to all Crossed Roller Bearings.
(2) CRBH is provided with an oil groove and two oil holes on the outer ring.
Remark n denotes the number of oil holes not exceeding 4. For one oil hole, number is not indicated.
When preparing multiple oil holes, please contact IKO.

Example 1 When the inner ring has 4 oil holes

CRBC 10020 / 4OH C1
4 Oil holes
In inner ring

Example 2 When the outer ring has a single oil hole

CRBC 10020 - OH C1
1 Oil hole
In outer ring

Operating Temperature Range

The operating temperature range for Crossed Roller Bearings is −20℃～+120℃. However, the maximum allowable temperature for types with separator and with seal is +110℃, and +100℃ when they are continuously operated.

Mounting

When the rigidity of the mounting parts is not sufficient, stress concentration will occur at the contact area between the rollers and raceways, and the bearing performance will be deteriorated significantly.
Therefore, it is necessary to carefully examine the rigidity of housing and the strength of fixing bolts when a large moment will be applied.
The shoulder height diameters (d_a and D_a) that are related to mounting should certainly satisfy the values shown in the dimension tables. When these dimensions are incorrect, deformations of inner and outer rings will occur and the bearing performance will be deteriorated remarkably.

1. For Mounting Holed Type High Rigidity Crossed Roller Bearing

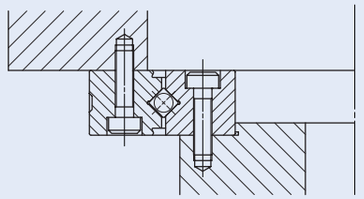


Fig.2 Mounting example -1
Mounting Holed Type High Rigidity Crossed Roller Bearing

❶Mounting Holed Type High Rigidity Crossed Roller Bearing can be mounted directly to the mounting surface by fixing bolts. (See Fig.3)

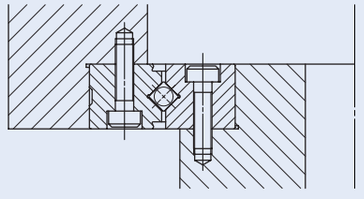


Fig.3 Mounting example -2
Mounting Holed Type High Rigidity Crossed Roller Bearing

❷If large number of radial load and/or moment is expected, it is recommended to prepare flange part. (See Fig.4)

❸Mounting Holed Type High Rigidity Crossed Roller Bearing has a plug for hole for inserting cylindrical rollers. When mounting the bearings, locate the plug at a position that is not included in the maximum loading zone. The plug location can be found by the pin that is at the side of the outer ring.

2. For other Crossed Roller Bearings

❶When the rigidity of the mounting parts is insufficient, stress concentration will occur at the contact area between the rollers and the raceways, and the bearing performance will be deteriorated significantly . Therefore, carefully examine the rigidity of housing and the strength of fixing bolts when a large moment is applied.

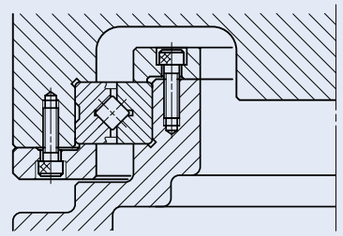


Fig. 4 Mounting example

❷The inner and outer rings should be securely fixed in the axial direction by using fixing plates, etc. Recommended thickness of the fixing plate is 1/2 or more of the bearing width B. The dimensions in the axial direction of the housing bore and the fixing plates should be determined to get a secure fixing while considering the dimension of bearing width which is given a minus tolerance.
❸The shoulder height diameters (d_a and D_a) that are related to mounting should satisfy the values shown in the dimension tables. When these dimensions are incorrect, deformations of inner and outer rings will occur and the bearing performance will be remarkably impaired.
❹The depth of the housing bore is recommended to be equal to or larger than the bearing width.
❺Separation prevention bolts for the outer ring are provided to prevent separation of two halves of the outer ring during transportation or mounting. When mounting, they should be loosened slightly.
❻High Rigidity Type Crossed Roller Bearings and Slim Type Crossed Roller Bearings have a plug for hole for inserting rollers. When mounting the bearings, locate the plug at a position that is not included in the maximum loading zone. The plug location can be found by the pin that is at the side of the outer ring.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CROSSED ROLLER BEARINGS

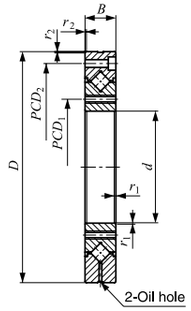
Mounting Holed Type High Rigidity Crossed Roller Bearings

Open type with separator

Two sides sealed type with separator



Shaft dia. 10 — 80mm



CRBF···AT



CRBF···ATUU



CRBF···A



CRBF···AUU

Shaft diameter mm	Identification number Open type	Identification number Two side sealed type	Weight (Ref.) kg	Boundary dim. mm d	D	B	$r_{1\min}$ (1)	$r_{2\min}$ (1)
10	CRBF 108 AT	CRBF 108 AT UU	0.12	10	52	8	0.3	0.3
20	CRBF 2012 AT	CRBF 2012 AT UU	0.31	20	70	12	0.3	0.3
25	CRBF 2512 AT	CRBF 2512 AT UU	0.40	25	80	12	0.6	0.6
35	CRBF 3515 AT	CRBF 3515 AT UU	0.66	35	95	15	0.6	0.6
55	CRBF 5515 AT	CRBF 5515 AT UU	0.96	55	120	15	0.6	0.6
80	CRBF 8022 AT	CRBF 8022 AT UU	2.63	80	165	22	0.6	1
	CRBF 8022 A	CRBF 8022 A UU	2.60					
	CRBF 8022 AD	CRBF 8022 AD UU						

Note(1) Minimum allowable single value of chamfer r_1 and r_2.

Remarks1. Outer ring has an oil groove and two oil holes.

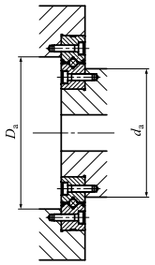
2. Open type is supplied without grease. Perform proper lubrication. Grease is pre-packed in two sides sealed type.

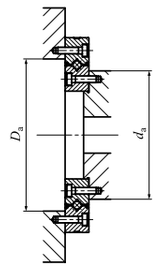


CRBF···AD



CRBF···ADUU





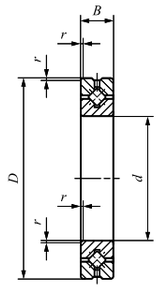
Mounting holes mm PCD_1	Inner ring Mounting holes	PCD_2	Outer ring Mounting holes	Mounting dimensions mm d_a	D_a	Basic dynamic load rating C N	Basic static load rating C_0 N
16	4-M3 through	42	6-ϕ 3.4 through ϕ 6.5 counter bore depth 3.3	24	31	2 910	2 430
28	6-M3 through	57	6-ϕ 3.4 through ϕ 6.5 counter bore depth 3.3	36.5	48.5	7 600	8 370
35	6-M3 through	67	6-ϕ 3.4 through ϕ 6.5 counter bore depth 3.3	46.5	58.5	8 610	10 600
45	8-M4 through	83	8-ϕ 4.5 through ϕ 8 counter bore depth 4.4	56	74	17 300	20 900
65	8-M5 through	105	8-ϕ 5.5 through ϕ 9.5 counter bore depth 5.4	76	94	20 100	27 700
97	10-M5 through	148	10-ϕ 5.5 through ϕ 9.5 counter bore depth 5.4	107	137	51 100	72 000
	10-ϕ 5.5 through ϕ 9.5 counter bore depth 5.4						

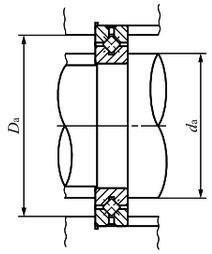
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

CROSSED ROLLER BEARINGS

High Rigidity Type Crossed Roller Bearings Open Type/With Separator







CRBH···A

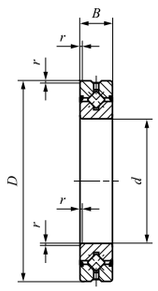
Shaft dia. 20 — 250mm

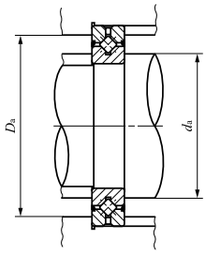
Shaft dia. mm	Identification number	Mass (Ref.) kg	Boundary dimensions mm d	D	B	r_{min} (1)	Mounting dimensions mm d_a	D_a	Basic dynamic load rating C N	Basic static load rating C_0 N
20	**CRBH 208 A**	0.04	20	36	8	0.3	24	31	2 910	2 430
25	**CRBH 258 A**	0.05	25	41	8	0.3	29	36	3 120	2 810
30	**CRBH 3010 A**	0.12	30	55	10	0.3	36.5	48.5	7 600	8 370
35	**CRBH 3510 A**	0.13	35	60	10	0.3	41.5	53.5	7 900	9 130
40	**CRBH 4010 A**	0.15	40	65	10	0.3	46.5	58.5	8 610	10 600
45	**CRBH 4510 A**	0.16	45	70	10	0.3	51.5	63.5	8 860	11 300
50	**CRBH 5013 A**	0.29	50	80	13	0.6	56	74	17 300	20 900
60	**CRBH 6013 A**	0.33	60	90	13	0.6	66	84	18 800	24 300
70	**CRBH 7013 A**	0.38	70	100	13	0.6	76	94	20 100	27 700
80	**CRBH 8016 A**	0.74	80	120	16	0.6	88	112	32 100	43 400
90	**CRBH 9016 A**	0.81	90	130	16	0.6	98	122	33 100	46 800
100	**CRBH 10020 A**	1.45	100	150	20	0.6	110	140	50 900	72 200
110	**CRBH 11020 A**	1.56	110	160	20	0.6	120	150	52 400	77 400
120	**CRBH 12025 A**	2.62	120	180	25	1	132	168	73 400	108 000
130	**CRBH 13025 A**	2.82	130	190	25	1	142	178	75 900	115 000
140	**CRBH 14025 A**	2.96	140	200	25	1	152	188	81 900	130 000
150	**CRBH 15025 A**	3.16	150	210	25	1	162	198	84 300	138 000
200	**CRBH 20025 A**	4.0	200	260	25	1	212	248	92 300	169 000
250	**CRBH 25025 A**	4.97	250	310	25	1.5	262	298	102 000	207 000

Note(1) Minimum allowable single value of chamfer dimension r
Remarks1. The outer ring has an oil groove and two oil holes.
2. Grease is not prepacked. Perform proper lubrication.

High Rigidity Type Crossed Roller Bearings Sealed Type/With Separator







CRBH···AUU

Shaft dia. 20 — 250mm

Shaft dia. mm	Identification number	Mass (Ref.) kg	Boundary dimensions mm d	D	B	r_{min} (1)	Mounting dimensions mm d_a	D_a	Basic dynamic load rating C N	Basic static load rating C_0 N
20	**CRBH 208 A UU**	0.04	20	36	8	0.3	24	31	2 910	2 430
25	**CRBH 258 A UU**	0.05	25	41	8	0.3	29	36	3 120	2 810
30	**CRBH 3010 A UU**	0.12	30	55	10	0.3	36.5	48.5	7 600	8 370
35	**CRBH 3510 A UU**	0.13	35	60	10	0.3	41.5	53.5	7 900	9 130
40	**CRBH 4010 A UU**	0.15	40	65	10	0.3	46.5	58.5	8 610	10 600
45	**CRBH 4510 A UU**	0.16	45	70	10	0.3	51.5	63.5	8 860	11 300
50	**CRBH 5013 A UU**	0.29	50	80	13	0.6	56	74	17 300	20 900
60	**CRBH 6013 A UU**	0.33	60	90	13	0.6	66	84	18 800	24 300
70	**CRBH 7013 A UU**	0.38	70	100	13	0.6	76	94	20 100	27 700
80	**CRBH 8016 A UU**	0.74	80	120	16	0.6	88	112	32 100	43 400
90	**CRBH 9016 A UU**	0.81	90	130	16	0.6	98	122	33 100	46 800
100	**CRBH 10020 A UU**	1.45	100	150	20	0.6	110	140	50 900	72 200
110	**CRBH 11020 A UU**	1.56	110	160	20	0.6	120	150	52 400	77 400
120	**CRBH 12025 A UU**	2.62	120	180	25	1	132	168	73 400	108 000
130	**CRBH 13025 A UU**	2.82	130	190	25	1	142	178	75 900	115 000
140	**CRBH 14025 A UU**	2.96	140	200	25	1	152	188	81 900	130 000
150	**CRBH 15025 A UU**	3.16	150	210	25	1	162	198	84 300	138 000
200	**CRBH 20025 A UU**	4.0	200	260	25	1	212	248	92 300	169 000
250	**CRBH 25025 A UU**	4.97	250	310	25	1.5	262	298	102 000	207 000

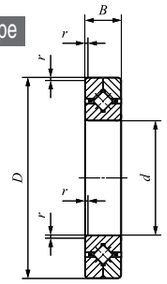
Note(1) Minimum allowable single value of chamfer dimension r
Remarks1. The outer ring has an oil groove and two oil holes.
2. Provided with prepacked grease.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

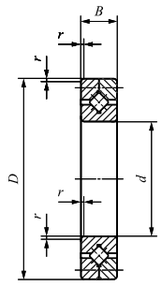
CROSSED ROLLER BEARINGS

Standard Type Crossed Roller Bearings **Open Type/With Cage** **Open Type/Full Complement Type**





CRBC



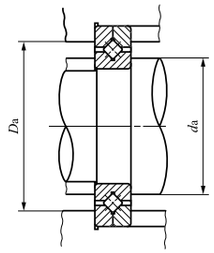
CRB

Shaft dia. 30 — 250mm

Shaft dia. mm	Identification number With Cage	Identification number Full complement	Mass (Ref.) kg	Boundary dimensions mm d	D	B	r_{min} (1)	Mounting dimensions mm d_a	D_a	CRBC Basic dynamic load rating C N	CRBC Basic static load rating C_0 N	CRB Basic dynamic load rating C N	CRB Basic static load rating C_0 N
30	CRBC 3010	CRB 3010	0.12	30	55	10	0.3	34	44	3 830	4 130	5 290	6 350
40	CRBC 4010	CRB 4010	0.15	40	65	10	0.3	44	54	4 280	5 140	5 980	8 040
50	CRBC 5013	CRB 5013	0.29	50	80	13	0.6	55	71	10 700	12 600	14 200	18 400
60	CRBC 6013	CRB 6013	0.33	60	90	13	0.6	64	81	11 600	14 600	15 400	21 500
70	CRBC 7013	CRB 7013	0.38	70	100	13	0.6	75	91	12 300	16 700	17 000	25 500
80	CRBC 8016	CRB 8016	0.74	80	120	16	0.6	86	107	18 200	25 500	24 300	37 500
90	CRBC 9016	CRB 9016	0.81	90	130	16	1	98	118	19 400	28 600	25 900	42 100
100	CRBC 10020	CRB 10020	1.45	100	150	20	1	108	134	31 500	45 100	39 400	61 100
110	CRBC 11020	CRB 11020	1.56	110	160	20	1	118	144	33 500	50 700	41 200	66 700
120	CRBC 12025	CRB 12025	2.62	120	180	25	1.5	132	164	47 700	70 500	59 900	95 400
130	CRBC 13025	CRB 13025	2.82	130	190	25	1.5	140	172	49 200	74 800	61 000	99 800
140	CRBC 14025	CRB 14025	2.96	140	200	25	1.5	151	183	50 700	79 200	64 100	108 000
150	CRBC 15025	CRB 15025	3.16	150	210	25	1.5	160	192	53 800	87 700	65 000	113 000
	CRBC 15030	CRB 15030	5.3	150	230	30	1.5	166	202	69 200	108 000	85 900	144 000
200	CRBC 20025	CRB 20025	4.0	200	260	25	2	208	239	60 200	110 000	75 300	148 000
	CRBC 20030	CRB 20030	6.7	200	280	30	2	218	262	108 000	178 000	133 000	234 000
	CRBC 20035	CRB 20035	9.58	200	295	35	2	221	274	137 000	215 000	168 000	282 000
250	CRBC 25025	CRB 25025	4.97	250	310	25	2.5	259	290	67 200	136 000	83 900	183 000
	CRBC 25030	CRB 25030	8.1	250	330	30	2.5	265	310	116 000	208 000	146 000	283 000
	CRBC 25040	CRB 25040	14.8	250	355	40	2.5	271	330	179 000	299 000	215 000	382 000

Note(1) Minimum allowable single value of chamfer dimension r

Remarks1. No oil hole is provided.
2. Grease is not prepacked. Perform proper lubrication.



1N=0.102kgf=0.2248lbs.
1mm=0.03937inch